



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO M.S TAMILVAANI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

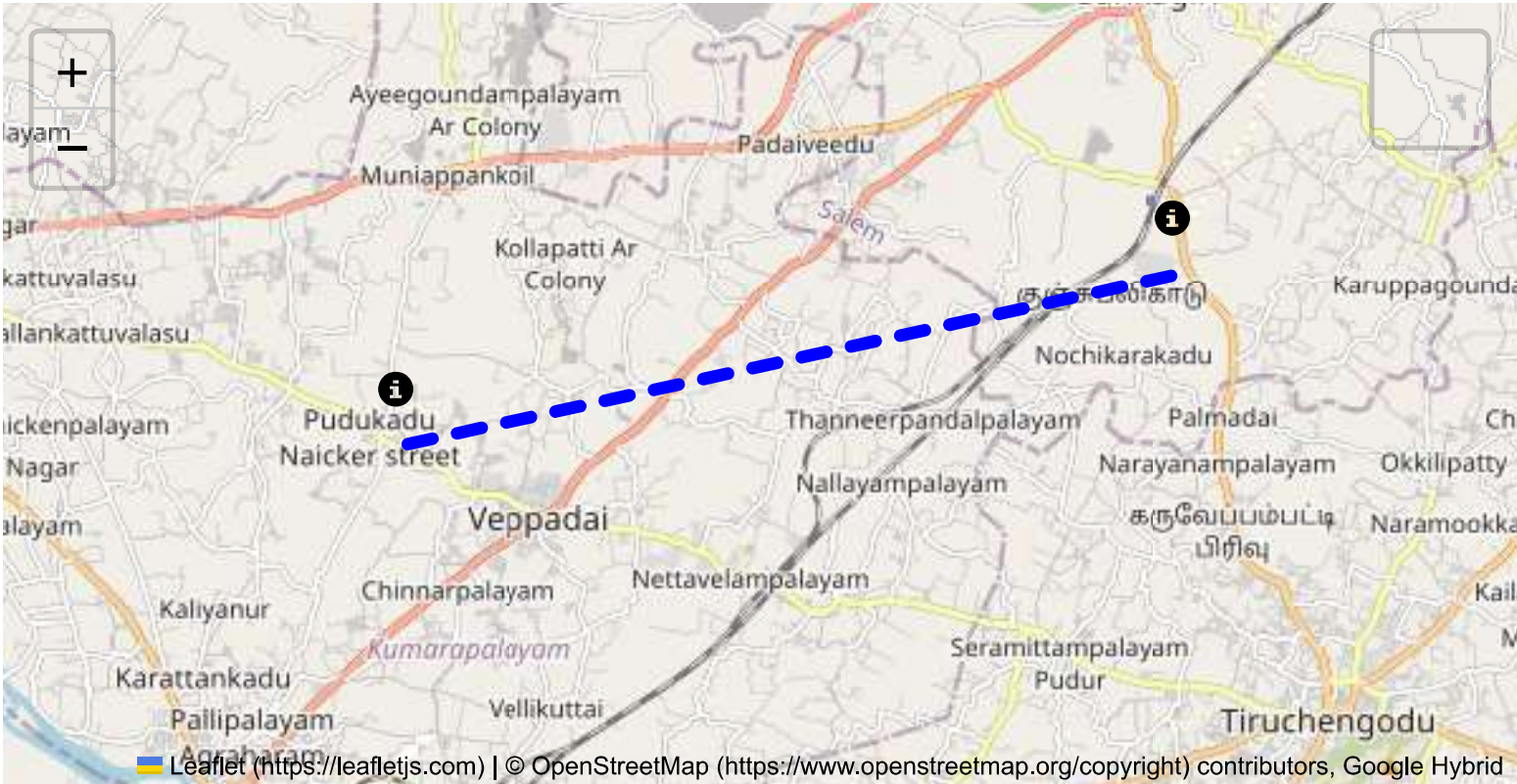
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

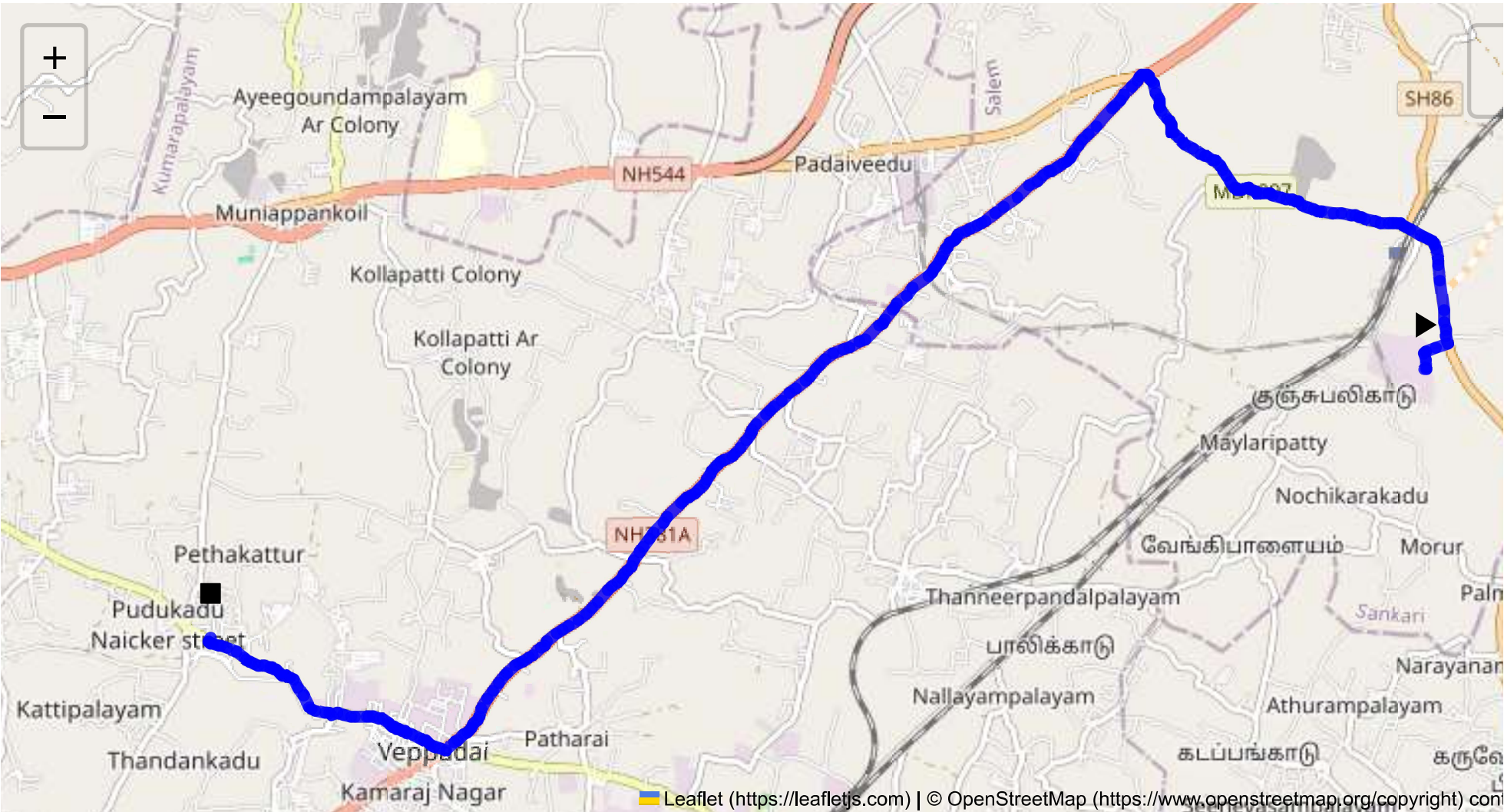
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 16.73 km
Estimated Duration: 0.4 hours
Adjusted Duration (Heavy Vehicle): 0.5 hours
Start: (11.4381, 77.8734)
End: (11.415223, 77.767738)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Ranganoor, Tamil Nadu

1. Overview of the Route Map

The route from CVQF+23W (Sangagiri) to CQ79+W69 (Ranganoor) primarily follows local roads in Tamil Nadu, covering approximately 16.73 kilometers. The travel path includes a mixture of rural roads and

possible small-town transit.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. The route could face:

- **Monsoon Season:** Heavy rains from June to September, potentially leading to water-logged roads.
- **Summer Heat:** High temperatures between March and May can cause road surface damage.
- **Fog:** Occasional fog in the early mornings during winters might impact visibility.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (7:00 AM - 9:00 AM) and evening (5:00 PM - 7:00 PM) might experience increased traffic due to local commuting patterns.
- **Congestion-prone Areas:** Town centers, particularly around Sangagiri and closer to Komarapalayam, may experience delays during peak hours due to market activity.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Generally mixed; rural stretches may feature narrow lanes with variable maintenance, whereas town roads may be better maintained but busier.
- **Infrastructure:** Limited street lighting in rural sectors may increase night-time hazard risks. Watch for bridges and culverts, potential chokepoints.

5. Suggestions for Alternative Routes for Emergencies

- **Rural Diversions:** Use agricultural roads in emergencies, though they may not support heavy truck traffic.
- **Highway Access:** Nearby NH-44 can provide a more stable alternate path in extreme conditions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits Required:** Ensure transport permits for hazardous materials are in place.
- **Time Restrictions:** Some areas may restrict large vehicles during peak hours.

7. Overview of Historical Incidents

- **Local Incidents:** Few publicly documented incidents primarily relate to minor accidents due to narrow roads.
- **Hazardous Material:** Historical data on major spills or accidents with hazardous materials is limited, indicating low prior incidents.

8. Environmental Considerations and Sensitive Areas

- **Water Bodies:** Small bodies of water near the route; risk of contamination if spill occurs.
- **Cultural Sites:** Proximity to small village centers requires caution to avoid disruptions.

9. Analysis of Communication Coverage

- **Coverage:** Generally good along the route due to proximity to towns but expect brief loss in more rural sections.
- **Dead Zones:** Possible in agricultural domains; ensure radio or alternative communication devices are available.

10. Estimated Emergency Response Times

- **Urban Areas:** 15-30 minutes due to proximity to services.
- **Rural Areas:** Could extend to 45-60 minutes depending on road conditions and distance.

12. Overall Summary of Risk Assessment

- **Overall Risk Level:** Moderate. While infrastructure and weather pose potential risks, the route is relatively short with adequate emergency access.
- **Primary Concerns:** Weather-related road impacts, narrow roads, traffic in town areas. Ensure vehicle maintenance, driver awareness, and communication readiness.
- **Recommendations:** Schedule trips outside peak hours, monitor weather forecasts, and prepare emergency protocols. Employ GPS navigation with updated traffic data for dynamic rerouting capabilities.

This comprehensive analysis intends to prepare the truck driver for potential challenges and ensure safe transportation of hazardous materials along this route in Tamil Nadu.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.46307, 77.84941	30 KM/Hr
5	Turn	Medium	11.46318, 77.84913	30 KM/Hr
6	Turn	Medium	11.46312, 77.84886	30 KM/Hr
7	Turn	High	11.40574, 77.78829	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	High	11.41493, 77.76777	15 KM/Hr

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46312, 77.84886



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40574, 77.78829



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41493, 77.76777